

A Diamond Obstruction for Low Fragmentability of Free Spaces over Nets

Run `fa_banach_001`, agent `agent_lane_17`

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Status

This is a partial result, not a full solution of the open problem in Remark 3.7 of Basset–Lancien–Prochazka [1]. It proves a broad high-index obstruction for genuine nets in Banach spaces. It does not compute the weak-fragmentability index of the canonical integer grid of ℓ_1 .

Source Problem

The source paper studies the weak-fragmentability index Φ and the dentability index of Lipschitz-free spaces over countable complete discrete metric spaces. On PDF page 10, Remark 3.7 asks for examples of nets in infinite-dimensional Banach spaces, or integer grids of infinite semi-normalized Schauder bases, for which the free space has weak-fragmentability index ω . The remark points to the canonical integer grid of ℓ_1 as a natural candidate.

$$\Psi(\mathcal{F}(\mathbb{N})) = \Psi(\mathcal{F}(\mathbb{Z}_{Haar})) = \omega.$$

Remark 3.7. Determining what are the possible values of $\Phi(\mathcal{F}(M))$ for M separable complete and purely 1-unrectifiable and computing $\Phi(\mathcal{F}(M))$ is a general interesting question. The results in [4] imply that this set of possible values is uncountable. One important subquestion is to determine, for which M we have that $\Phi(\mathcal{F}(M)) = \omega$. Let us mention in this direction that it follows from results in [2] and [7] that if M is a proper (i.e. its closed balls are compact) p-1-u metric space, then $\mathcal{F}(M)$ is isometrically a dual space and its norm is asymptotically uniformly convex for the corresponding weak* topology. In particular $\mathcal{F}(M)$ is asymptotically uniformly convex and thus $\Phi(\mathcal{F}(M)) \leq \omega$.

We do not know any example of a net in an infinite dimensional Banach space or of an integer grid of an infinite semi-normalized Schauder basis such that the fragmentability index of its free space is equal to ω . In view of the results from [25] and [17], the integer grid of the canonical basis of ℓ_1 is a natural candidate.

For more information on this problem, see the paper [1] and the references therein.

Idea of the Proof

The source paper already proves the two index estimates that matter here. First, every uniformly discrete metric space M satisfies $\Phi(\mathcal{F}(M)) \leq \omega^2$; if M is infinite, only ω and ω^2 can occur below this bound. Second, if the countably branching diamond graphs $(D_n^\omega)_{n \in \mathbb{N}}$ embed into M with one uniform distortion bound, then $\Phi(\mathcal{F}(M)) \geq \omega^2$.

Thus it is enough to know when the diamond family survives inside a net. A net projection changes distances by an additive constant. In a Banach space, however, a bi-Lipschitz copy can first be dilated by an arbitrarily large scalar. After that dilation, nearest-net projection only perturbs distances by a small relative amount. Since graph distances in D_n^ω are integer-valued and at least one between distinct vertices, one scale absorbs the additive error for all pairs and all n .

Definitions

An (a, ρ) -net in a Banach space X means a subset $N \subset X$ such that $\|x - y\| \geq a$ for distinct $x, y \in N$, and every point of X lies within distance ρ of some point of N . The countably branching diamond graphs D_n^ω are used with their graph metrics d_n , normalized so that the shortest nonzero distance is 1.

A family (D_n^ω, d_n) embeds with equal distortion into a metric space M if there is a constant C such that each D_n^ω admits a bi-Lipschitz embedding into M with distortion at most C .

Lemma 1 (scaled nearest-net projection). *Let X be a Banach space and let N be an (a, ρ) -net in X . Suppose that the family (D_n^ω, d_n) embeds with equal distortion into X . Then (D_n^ω, d_n) embeds with equal distortion into N .*

Proof. After rescaling the individual embeddings, choose $L \geq 1$ and maps $f_n : D_n^\omega \rightarrow X$ such that for every $u, v \in D_n^\omega$,

$$d_n(u, v) \leq \|f_n(u) - f_n(v)\| \leq Ld_n(u, v).$$

Fix a scalar $R \geq \max\{1, 4\rho\}$. For each vertex $v \in D_n^\omega$, choose $g_n(v) \in N$ with

$$\|g_n(v) - Rf_n(v)\| \leq \rho.$$

If $u \neq v$, then $d_n(u, v) \geq 1$, so

$$\|g_n(u) - g_n(v)\| \geq R\|f_n(u) - f_n(v)\| - 2\rho \geq Rd_n(u, v) - 2\rho \geq \frac{R}{2}d_n(u, v).$$

Also,

$$\|g_n(u) - g_n(v)\| \leq R\|f_n(u) - f_n(v)\| + 2\rho \leq RLd_n(u, v) + 2\rho \leq R\left(L + \frac{1}{2}\right)d_n(u, v).$$

Thus each g_n is bi-Lipschitz, with distortion at most $2L + 1$, independently of n . This is the required equal-distortion embedding into N . $\square$

Theorem 1 (net diamond obstruction). *Let X be a Banach space and let N be an (a, ρ) -net in X . If the countably branching diamond graphs $(D_n^\omega)_{n \in \mathbb{N}}$ embed with equal distortion into X , then*

$$\Phi(\mathcal{F}(N)) = \omega^2.$$

Proof. By the lemma, the family $(D_n^\omega)_{n \in \mathbb{N}}$ embeds with equal distortion into N . Proposition 3.2 of [1] then yields $\Phi(\mathcal{F}(N)) \geq \omega^2$.

Since N is a net, it is uniformly discrete. Proposition 3.1 of [1] gives $\Phi(\mathcal{F}(N)) \leq \omega^2$. The two estimates coincide, so $\Phi(\mathcal{F}(N)) = \omega^2$. $\square$

Corollary 1. *Let X be a Banach space containing a linear isomorphic copy of a Banach space into which the family (D_n^ω) embeds with equal distortion. Then every net N in X satisfies $\Phi(\mathcal{F}(N)) = \omega^2$. In particular, this applies to Banach spaces containing an isomorphic copy of c_0 , and to spaces containing any of the L_1 diamond models cited in the source paper.*

Proof. Compose the uniform diamond embeddings with the linear isomorphic inclusion into X . This changes the common distortion bound by at most the Banach space isomorphism constant. The theorem applies. $\square$

Limitations

This result does not decide whether any integer grid of an infinite semi-normalized Schauder basis has free-space weak-fragmentability index ω . In particular, it does not compute $\Phi(\mathcal{F}(\mathbb{Z}_{\ell_1}))$. The proof uses coarse density of a genuine net in an ambient Banach space; an integer grid such as the canonical grid of ℓ_1 need not be a net in this sense.

The result should therefore be read as a large-family exclusion theorem: whenever the ambient Banach space already contains uniformly distorted countably branching diamonds, no genuine net in that space can provide the low-index example sought in Remark 3.7.

Verification and Novelty Check

The proof is analytic and uses no finite search. The evidence crop was generated by

```
conda run --no-capture-output -n sandbox python code/make_open_problem_crop.py.
```

Bounded duplicate and novelty checks were performed on 2026-06-20. The local run indexes were searched for arXiv:2505.19891, weak-fragmentability index, countably branching diamonds, the integer grid of ℓ_1 , and nearby net keywords. Web searches for the source arXiv id and for combinations of *weak-fragmentability index*, *Lipschitz-free space*, *integer grid*, ℓ_1 , and *countably branching diamond* found the source paper but no later computation of the ℓ_1 grid and no duplicate of this abstract net-obstruction packet.

Human Review Recommendation

Review as a likely valid partial result. The central point to check is the scaled nearest-net projection lemma, especially the use of the graph-metric normalization $d_n(u, v) \geq 1$ for distinct vertices. If accepted, the packet should be treated as an obstruction/family-classification result, not as a solution of the source paper’s proposed $\mathbb{Z}_{\ell_1}$ candidate.

References

- [1] E. Basset, G. Lancien, and A. Prochazka, *Diversity of Lipschitz-free spaces over countable complete discrete metric spaces*, arXiv:2505.19891, 2025.